Jânio Anselmo, M.Sc.

PhD Candidate in Electrical Engineering (Biomedical Engineering), UFSC
Electrical Engineer – CREA-SC 074576-3

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Summary

PhD candidate in Electrical Engineering at the Federal University of Santa Catarina (UFSC), Brazil, with a M.Sc. in the same program (Biomedical Engineering concentration). Doctoral research covers mathematical modeling and experimental validation of pulsed electric fields (PEF) in continuous flow, combining numerical modeling, laboratory instrumentation and parameter estimation against measured data.

Co-founder and R&D Director of ENSA Tecnologia since 2009, leading a team of four across biomedical signal acquisition, environmental sensing, industrial automation and pattern classification, and named inventor on two Brazilian patent applications. Published peer-reviewed work on numerical modeling of electroporation; peer reviewer for the American Medical Informatics Association (AMIA), member of five undergraduate thesis committees between 2014 and 2026, and teaching assistant for the Transducers Laboratory course at UFSC — work that consists of explaining technical concepts precisely and judging technical output against explicit criteria.

Education

Ph.D. Candidate, Electrical Engineering — Biomedical Engineering 2023–2027
Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil
Graduate Program in Electrical Engineering (PPGEEL). Thesis: *Pulsed electric fields in continuous flow: effective electric dose modeling and experimental validation*. Advisor: Prof. Daniela Ota Hisayasu Suzuki, Ph.D.; co-advisor: Prof. Guilherme Brasil Pintarelli, Ph.D.

M.Sc., Electrical Engineering — Biomedical Engineering 2012–2014
Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil
Dissertation: *Single-Cell Electroporation Study* — numerical study of the electrical and mechanical effects of capillary-electrode electroporation on the cell membrane.

B.Eng., Electrical Engineering — Telematics 2006–2011
University of Southern Santa Catarina (UNISUL) · Palhoça, Brazil
Electives in power electronics and computer networks.

Also: Postgraduate Specialization in Business Management, FMP (2015–2018) · B.Sc. Technology in Telecommunications Systems, IFSC (2008–2015) · Technical Diploma in Telecommunications, IFSC (1997–2001).

Research Experience

Doctoral Researcher — Pulsed Electric Fields in Continuous Flow 2023–2027
Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

- Developed an effective electric dose model for yeast (*Saccharomyces cerevisiae*) inactivation under continuous-flow PEF treatment, keeping pulse repetition rate and residence time as separate factors.
- Built and instrumented the experimental bench, acquiring oscilloscope voltage and current waveforms, temperature records and conductivity measurements as the validation set; then estimated the model parameters and assessed them out of sample, with leave-one-out validation and bootstrap confidence intervals for the fitted coefficients.
- Wrote the resulting manuscript in LaTeX under the IEEE Access template, in both English and Portuguese.

Master's Researcher — Single-Cell Electroporation 2012–2014
Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

Numerical study of electroporation of isolated biological cells through a stretched capillary electrode, covering the electrical and mechanical effects on the cell membrane. Advisors: Prof. Daniela O. H. Suzuki, Ph.D. and Prof. Jefferson L. B. Marques, Ph.D. In parallel, developed low-cost hardware for heartbeat and cardiac audio monitoring in the *Smart Medical Devices* course with Prof. Mohamad Sawan, P.Eng., Ph.D. (2013).

Publications

- **Anselmo, J.**, Suzuki, D. O. H., Pintarelli, G. B., Antônio Júnior, A. de C., Silva, J. R. da. *Mathematical Modeling and Experimental Assessment of Pulsed Electric Fields in Continuous Flow for Saccharomyces cerevisiae*. Submitted, under review, 2026.
- Suzuki, D. O. H., **Anselmo, J.**, de Oliveira, K. D., Freytag, J. O., Rangel, M. M. M., Marques, J. L. B., Ramos, A. *Numerical Model of Dog Mast Cell Tumor Treated by Electrochemotherapy*. Artificial Organs, 2014.
- Fronza, C. F., **Anselmo, J.**, Marques, J. L. B., Pintarelli, G. B., Castro, A. de. *A Blower-Like Approach to Predict the Effectiveness of Vaccines in a TB Dynamic*. International Journal of Engineering Research and Applications, v. 4, n. 6, pp. 233–238, 2014.
- Fronza, C. F., **Anselmo, J.**, Marques, J. L. B. *In Silico System for Early Diagnosis of Diabetes Mellitus Complications by Using Dynamic Pupillometry and Heart Rate Variability*. BIOMAT 2014 — International Symposium on Mathematical and Computational Biology, Poznan, Poland, 2014.
- **Anselmo, J.**, Suzuki, D. O. H., Marques, J. L. B. *Numerical Studies of the Influential Parameters of Electroporation Through Stretched Microcapillaries*. XXIV Brazilian Congress on Biomedical Engineering (CBEB), Uberlândia, Brazil, 2014.

Patents and Registrations

- **Anselmo, J.**; Pintarelli, G. B. *Resonant-excitation-based electronic electroporation system* (ElectroRES-100A). Brazil, 2026. Invention patent application filed through SINOVA, the UFSC innovation agency (UFSC code 2026032378000383). The treated biological medium is functionally part of the electrical circuit, enabling dynamic impedance monitoring and adaptive control of the electroporation process.
- **Anselmo, J.** *Electronic process using ultrasound technology for real-time volumetric measurement in liquid waste trucks*. Brazil, 2025. Invention patent, application no. BR 10 2025 001160 3, filed 22 January 2025 with INPI (Brazilian National Institute of Industrial Property). Funded by ENSA Tecnologia.
- **Anselmo, J.**; Anselmo, T. *Trademark registration for ENSA Tecnologia products and services*. Brazil, 2016. Registered trademark no. 925500895, INPI.

Peer Review and Teaching

Teaching Assistantship — Transducers Laboratory 2025, 2012
Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

Electrical and Electronic Engineering programs; two separate assistantships.

Article Reviewer 2013–2014
American Medical Informatics Association (AMIA)
Reviewed submitted manuscripts against the association's review criteria; three reviews completed in 2014.

Undergraduate Thesis Committee Member — five committees 2014–2026
UFSC and IFSC · Brazil
Theses examined: *Contact impedance sensing in electroporation: quantifying the effects of conductive gel anomalies in electrochemotherapy* (Electrical Engineering) and *Safety system design for a hydraulic press brake under NR-12: a retrofit case study based on ISO 12100 and ISO 13849* (Control and Automation Engineering), both UFSC, 2026; *Analysis of electrical, geometrical and chemical parameters in the application of nanosecond electric fields to biological cells* and *In vitro electroporation study: numerical simulation and experiments*, both UFSC Electronic Engineering, 2017; *Epileptic seizure detection from electroencephalogram signals using the Wavelet transform*, IFSC Telecommunications Systems, 2014.

Instructor — Biomedical Signal Processing 2013
V Minicourse on Biomedical Engineering in Practice · Brazil
Designed and taught a short course on biomedical signal processing.

Organizer 2012
IV Minicourse on Biomedical Engineering in Practice · Brazil

Professional Experience

Co-founder and R&D Director 2009–Present
ENSA Tecnologia · Palhoça, Brazil
Lead the company's research and development function with a team of four, owning product concept and electronics design end to end — from microcontroller hardware to embedded algorithms — across healthcare, industrial automation, environmental monitoring and IoT. Recent work includes the ultrasound-based real-time volumetric measurement process for liquid waste trucks, filed as a patent in 2025 (see *Patents and Registrations*).

Selected products designed in this role:

- **ENSABiofeedback-100A** (2021) — microcontroller-based home-care device acquiring multiple physiological signals: photoplethysmography (PPG), electromyography (EMG) and galvanic skin response (GSR), for biofeedback assessment and training.
- **ENSAPlate** (2017) — optical character recognition and pattern classification using artificial neural networks, applied to vehicle license plate recognition for fuel pump automation.
- **ENSAIoT-200A** (2017) and **ENSAIoT-100A** (2016) — IoT monitoring of volume, pressure, water flow and temperature with real-time SaaS transmission, built on an open-source microcontroller board with analog and digital I/O, Bluetooth, Wi-Fi, RS-232/485, LCD, touch keypad and relay outputs.
- **ENSAMed-200S** (2019) and **ENSAMed-100B** (2011) — environmental sensing and volumetric measurement for fuel station pump sumps and tank interstitial spaces.
- Earlier: biometric access control over Ethernet and GSM/GPRS (2015), RFID-operated retail entertainment platform (2012) and a monitored residential alarm panel (2009).

Technical Skills

Programming and modeling: Python, Matlab, Octave, C/C++, Java; COMSOL Multiphysics, PSpice, Proteus

Embedded and electronics CAD: Microchip PIC, Atmel AVR, Espressif ESP; Altium Designer, KiCAD, Eagle

Instrumentation: Oscilloscope data acquisition, impedance spectroscopy, conductivity measurement

Scientific writing: LaTeX (IEEE templates), technical documentation

Open-source work: github.com/janioanselmo — Python tools for oscilloscope waveform analysis, impedance spectroscopy and literature search

Languages

Portuguese: Native · **English:** Intermediate · **Spanish:** Basic